Dominik Stiftinger-Lang

Senior Backend & AI Systems Engineer
Go, Python, Machine Learning, AI Agents

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Profile summary

Senior engineer, mathematician and founder with nine years in financial systems. Built production ML and distributed Go systems, published neural-network research, and led companies, products and projects.

Contracting via ChainBrain FlexCo
Remote from Graz, Austria

Expertise

- Go & Distributed Systems
- Trading & Risk Systems
- PostgreSQL & APIs
- AI-Assisted Engineering
- Machine Learning & RL
- Company & Project Management

Engagement

Available
1 September 2026

Capacity
30-40 hours per week

Standard remote rate
EUR 120/hour excluding VAT

Contract model
B2B via Austrian FlexCo

Languages

German	Native
English	Fluent

Selected Projects

2026 - present
ChainBrain FlexCo

AI-Assisted Software Delivery and Quality Control

AI-Assisted Engineering Lead / Systems Architect

- Claude Code and Codex workflows, multi-agent orchestration, reusable skills, and MCP-based tools.
- Evidence-based branch verdicts for AI-generated changes in a financially sensitive Go monorepo.

2022 - present
ChainBrain FlexCo

Distributed Portfolio Management and Trading Platform

Co-Founder, CEO, Lead Backend Engineer and Architect

- Product ownership and Scrum-based delivery alongside hands-on architecture and implementation.
- Production Go platform with Temporal workflows, PostgreSQL, gRPC, OpenAPI, and live client capital.
- Architecture spanning portfolio construction, execution, reconciliation, reporting, and regulated boundaries.

2020 - 2022
Nerox / TU Graz / PhD

Production ML and Reinforcement-Learning Research

Quantitative Researcher / ML Engineer

- Built the 2020-2021 Apex portfolio ML pipeline: data and feature engineering, TensorFlow/Keras training, automated tuning and backtesting, and staged live trading.
- Published spiking-neural-network research in 2021 and continued reinforcement-learning research for DeFi portfolio constraints from 2022.

Technical Focus

- Go and Python
- Distributed Systems
- PostgreSQL and Temporal
- Docker and GitLab CI
- Company and Project Management
- Claude Code and Codex
- MCP and AI Agents
- REST, gRPC and OpenAPI
- TensorFlow, PyTorch and RL
- Product Ownership and Scrum

Core Technologies

Languages & Machine Learning
Go / Python
SQL
TensorFlow / PyTorch
Machine learning / Reinforcement learning

Architecture
Distributed systems
Temporal workflows
REST, gRPC, OpenAPI
WebSocket

Management & Delivery
Company leadership
Project planning
Product ownership
Scrum

Project History

ChainBrain FlexCo, Graz

2026 - present

AI-Assisted Engineering Lead / Systems Architect

- Designed multi-agent workflows with bounded roles for implementation, codebase research, test generation, security and financial-flow audits, and independent verification.
- Standardized Claude Code and OpenAI Codex through reusable skills, task specifications, ownership rules, structured handoffs, and persistent engineering knowledge.
- Integrated MCP-based tools for knowledge retrieval and operational workflows, including a Python knowledge service over structured project documentation.
- Designed an evidence-based QC harness with static and unit checks, debt and reuse analysis, traceable findings, and explicit branch verdicts.

Technologies: Claude Code, OpenAI Codex, MCP, Go, Python, Bash, GitLab CI, Linux

ChainBrain FlexCo, Graz

2022 - present

Co-Founder, CEO, Lead Backend Engineer and Architect

- Led product and engineering delivery as Product Owner, translating product and stakeholder goals into backlog priorities, sprint scope, architecture decisions, and coordinated delivery.
- Wrote the project plan for an FFG-funded R&D project and managed delivery through final reporting.
- Architected and implemented distributed Go services with Temporal workflows, PostgreSQL persistence, gRPC and OpenAPI interfaces, and idempotent financial processing.
- Designed service and data boundaries for portfolio construction, account state, execution, reconciliation, reporting, and client-facing APIs.
- Implemented Docker and GitLab CI delivery, Vault and Consul infrastructure, monitoring, incident response, and controlled secrets handling.
- Defined a MiCA-aware architecture with explicit custody, authorization, customer-controlled execution, hosted analytics, and audit boundaries.

Technologies: Go, Python, PostgreSQL, Temporal, gRPC, OpenAPI, REST, Hasura, Docker, GitLab CI, Vault, Consul

ChainBrain FlexCo / confidential professional asset-management deployment

2024 - 2026

Lead Engineer, Multi-Venue Order Management and Execution

- Built REST and WebSocket integrations for Binance, Kraken, Bitfinex, Poloniex, and OKX, including venue-specific order semantics and recovery behaviour.
- Developed order-lifecycle state machines, fill and balance reconciliation, exposure controls, a B-tree orderbook, and execution-quality measurement.
- Operated production spot and futures rebalancing with sub-1 bps median implementation shortfall and no exposure-limit breaches.

Technologies: Go, PostgreSQL, Temporal, REST, WebSocket, gRPC, Docker

:: Earlier Experience

Nerox GmbH, Graz

2019 - 2026

Co-Founder, Head of Quantitative Finance, later Chief Risk Officer

- Co-designed and built Apex (2020-2021), an autonomous portfolio-management pipeline spanning real-time Go/WebSocket market-data collection, PostgreSQL storage, model deployment, and live trading.
- Implemented R/SQL/Python feature engineering and TensorFlow/Keras training with a ResNet-derived allocation model, portfolio-value objective, automated hyperparameter tuning, validation, backtesting, and staged capital rollout.
- Owned quantitative design, validation, deployment, and communication with business stakeholders.
- Served as Scrum Master until the role was handed over in 2023, then focused fully on Product Owner responsibility.

Technologies: Python, TensorFlow, Keras, R, PostgreSQL, SQL, Go, WebSocket, machine learning, reinforcement learning

Nerox GmbH, Graz

2017 - 2019

Software Developer

- Implemented low-latency trading strategies and Go exchange adapters using REST and WebSocket market-data and order APIs.

Graz University of Technology

2020 - 2021

Research Assistant, Institute for Theoretical Computer Science

- Developed and evaluated spiking-neural-network models in computational neuroscience; co-authored the resulting 2021 publication with Christoph Stoeckl and Wolfgang Maass.

:: Education

In progress — **University of Graz**
PhD Mathematics
Portfolio optimization under DeFi staking and lock-up constraints. Developed reinforcement-learning experiments from 2022, including a Gym environment and a TensorFlow/Keras actor-critic implementation.

2017 - 2019 — **Graz University of Technology**
MSc Mathematics
Numerical methods for partial differential equations

2013 - 2017 — **Graz University of Technology**
BSc Mathematics, Technomathematics

:: Selected Publications

2025 — **A Multi-Factor Investment Framework for Crypto Asset Management**
Crypto Valley Association Research Journal. Best-rated paper of 12 submissions.

2021 — **Structure induces computational function in networks with diverse types of spiking neurons**
Stoeckl, Lang, Maass. bioRxiv 10.1101/2021.05.18.444689.

2019 — **A Study of Boundary Element Methods for Electrostatic Transmission Problems**
MSc thesis, Graz University of Technology.